

1 Magnetic field on circular drop

Maxwells equations are:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = 0 \quad (1.1)$$

We introduce a scalar potential ψ such that $\mathbf{H} = -\nabla\psi$ hence we have

$$\nabla^2\psi = 0. \quad (1.2)$$

The magnetic induction $\mathbf{B}$ is defined as

$$\mathbf{B} = \begin{cases} \mu_d \mathbf{H} & \text{in drop} \\ \mu_0 \mathbf{H} & \text{in matrix} \end{cases} \quad (1.3)$$

The boundary conditions are $[\mathbf{B} \cdot \mathbf{n}] = 0$ and $[\mathbf{H} \cdot \mathbf{t}] = 0$. The far field condition will be a uniform field $-H_\infty \mathbf{j}$ so we need

$$\mathbf{H} = -H_\infty(\sin \theta \mathbf{e}_r + \cos \theta \mathbf{e}_\theta) = -\nabla\psi \quad (1.4)$$

Solutions to Laplace's equation are $\psi(r, \theta) = R(r)T(\theta)$ where $R(r) = r^\lambda + r^{-\lambda}$, $T(\theta) = \sin(\lambda\theta) + \cos(\lambda\theta)$. To satisfy the far field we need $\lambda = 1$ and $T(\theta) = \beta \sin \theta$. Solutions to this are

$$\psi(r, \theta) = \begin{cases} Ar \sin \theta & r \leq a \\ (H_\infty r + B/r) \sin \theta & r > a \end{cases} \quad (1.5)$$

where

$$A = \frac{2\mu_0}{\mu_0 + \mu_d} H_\infty, \quad B = \frac{\mu_0 - \mu_d}{\mu_0 + \mu_d} a^2 H_\infty$$

Hence

$$\mathbf{H} = \begin{cases} -A(\sin \theta \mathbf{e}_r + \cos \theta \mathbf{e}_\theta) & r \leq a \\ (-H_\infty + \frac{B}{r^2}) \sin \theta \mathbf{e}_r - (H_\infty + \frac{B}{r^2}) \cos \theta \mathbf{e}_\theta & r > a \end{cases} \quad (1.6)$$

2 Fluid motion in magnetic field

We introduce a stream function ϕ such that

$$u = \frac{1}{r} \frac{\partial \phi}{\partial \theta}, \quad v = -\frac{\partial \phi}{\partial r} \quad (2.1)$$

The biharmonic equation is

$$\nabla^4 \phi = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right)^2 \phi = 0 \quad (2.2)$$

The boundary conditions to satisfy are a far field $\mathbf{u} = 0$ as $|r| \rightarrow \infty$ outside the drop. There's dynamic stress conditions on the surface of the drop.

We look at the dynamic stress boundary conditions. They are as follows:

$$[\mathbf{n} \cdot \mathbf{T}^f \cdot \mathbf{t}] = 0 \quad (2.3)$$

$$[\mathbf{n} \cdot \mathbf{T}^f \cdot \mathbf{n}] + [\mathbf{n} \cdot \mathbf{T}^m \cdot \mathbf{n}] = \gamma \kappa \quad (2.4)$$

where

$$\mathbf{T}_i^f = -p_i \mathbf{I} + \frac{\eta_i}{2} (\nabla \mathbf{u}_i + (\nabla \mathbf{u}_i)^T) \quad (2.5)$$

$$\mathbf{T}_i^m = \mu_i \left(\mathbf{H}_i \mathbf{H}_i^T - \frac{1}{2} |\mathbf{H}_i|^2 \mathbf{I} \right) \quad (2.6)$$

or in matrix form:

$$\mathbf{T}_i^f = \begin{bmatrix} -p_i + \eta_i \frac{\partial u}{\partial r} & \frac{\eta_i}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right) \\ \frac{\eta_i}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right) & -p_i + \eta_i \left(\frac{1}{r} \frac{\partial v}{\partial \theta} + \frac{u}{r} \right) \end{bmatrix}$$

$$\mathbf{T}_i^m = \mu_i \begin{bmatrix} \frac{\partial \psi_i}{\partial r}^2 - \frac{1}{2} \left(\frac{\partial \psi_i}{\partial r}^2 + \frac{1}{r^2} \frac{\partial \psi_i}{\partial \theta}^2 \right) & \frac{1}{r} \frac{\partial \psi_i}{\partial \theta} \frac{\partial \psi_i}{\partial r} \\ \frac{1}{r} \frac{\partial \psi_i}{\partial \theta} \frac{\partial \psi_i}{\partial r} & \frac{1}{r^2} \frac{\partial \psi_i}{\partial \theta}^2 - \frac{1}{2} \left(\frac{\partial \psi_i}{\partial r}^2 + \frac{1}{r^2} \frac{\partial \psi_i}{\partial \theta}^2 \right) \end{bmatrix}$$

$$(\mathbf{e}_r, 0) \cdot \mathbf{T}^f \cdot \mathbf{e}_\theta = \left(\left[-p_i + \eta_i \frac{\partial u}{\partial r} \right] \mathbf{e}_r + \left[\frac{\eta_i}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right) \right] \mathbf{e}_\theta \right) \cdot \mathbf{e}_\theta$$

$$= \frac{\eta_i}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right)$$

$$\begin{aligned}
(\mathbf{e}_r, 0) \cdot \mathbf{T}^f \cdot \mathbf{e}_r &= \left(\left[-p_i + \eta_i \frac{\partial u}{\partial r} \right] \mathbf{e}_r + \left[\frac{\eta_i}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right) \right] \mathbf{e}_\theta \right) \cdot \mathbf{e}_r \\
&= -p_i + \eta_i \frac{\partial u}{\partial r}
\end{aligned}$$

$$\begin{aligned}
(\mathbf{e}_r, 0) \cdot \mathbf{T}^m \cdot \mathbf{e}_r &= \mu_i \left[\frac{\partial \psi_i^2}{\partial r} - \frac{1}{2} \left(\frac{\partial \psi_i^2}{\partial r} + \frac{1}{r^2} \frac{\partial \psi_i^2}{\partial \theta} \right) \right] \mathbf{e}_r + \mu_i \left[\frac{1}{r} \frac{\partial \psi_i}{\partial \theta} \frac{\partial \psi_i}{\partial r} \right] \mathbf{e}_r \\
&= \mu_i \frac{\psi_i^2}{\partial r} - \frac{\mu_i}{2} \left(\frac{\partial \psi_i^2}{\partial r} + \frac{1}{r^2} \frac{\partial \psi^2}{\partial \theta} \right)
\end{aligned}$$

Hence the boundary conditions are:

$$\left[\frac{\eta}{2} \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \theta} \right) \right] = 0 \quad (2.7)$$

$$\left[-p + \eta \frac{\partial u}{\partial r} \right] + \left[\frac{\mu}{2} \left(\left(\frac{\partial \psi}{\partial r} \right)^2 - \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta} \right)^2 \right) \right] = \gamma \kappa \quad (2.8)$$

Equation (2.7) can be written in terms of the fluid stream function ϕ as

$$\left[\eta \left(r \frac{\partial}{\partial r} \left(-\frac{1}{r} \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} \right) \right] = 0 \quad (2.9)$$

We also have continuity of fluid velocity so $[\mathbf{u}] = 0 \Rightarrow \left[\frac{1}{r} \frac{\partial \phi}{\partial \theta} \mathbf{e}_r - \frac{\partial \phi}{\partial r} \mathbf{e}_\theta \right] = 0$. Thus we get

$$\left[\frac{\partial \phi}{\partial \theta} \right] = 0, \quad \left[\frac{\partial \phi}{\partial r} \right] = 0 \quad (2.10)$$

Equation (2.8) is a modification of the magnetic Young-Laplace equation but with fluid motion. We demonstrate this by recalling $\mathbf{H} = -\nabla \psi = -\left(\frac{\partial \psi}{\partial r}, \frac{1}{r} \frac{\partial \psi}{\partial \theta} \right)$ so $\left(\frac{\partial \psi}{\partial r} \right)^2 = (-\mathbf{H} \cdot \mathbf{n})^2$ and $\left(\frac{1}{r} \frac{\partial \psi}{\partial \theta} \right)^2 = (-\mathbf{H} \cdot \mathbf{t})^2$. Hence we write (2.8) as

$$\left[-p + \eta \frac{\partial u}{\partial r} \right] + \left[\frac{\mu}{2} \left((\mathbf{H} \cdot \mathbf{n})^2 - (\mathbf{H} \cdot \mathbf{t})^2 \right) \right] = \gamma \kappa \quad (2.11)$$

The second jump term in equation 2 is

$$\left[\frac{\mu}{2} \left((\mathbf{H} \cdot \mathbf{n})^2 - (\mathbf{H} \cdot \mathbf{t})^2 \right) \right] = \frac{1}{2} \left(\mu_1 A^2 - \mu_2 H_\infty^2 - \frac{\mu_2 B^2}{a^4} \right) \cos(2\theta) - \frac{\mu_2 H_\infty B}{a^2} \quad (2.12)$$

Which after a bit of algebra simplifies to

$$\left[\frac{\mu}{2} \left((\mathbf{H} \cdot \mathbf{n})^2 - (\mathbf{H} \cdot \mathbf{t})^2 \right) \right] = -\frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)}{\mu_2 + \mu_1} \left(\frac{\mu_2 - \mu_1}{\mu_2 + \mu_1} \cos(2\theta) + 1 \right) \quad (2.13)$$

We need to think about what form ϕ takes. From the boundary condition (2.11) and (2.13) we deduce that u should be similar to $\cos(2\theta)$ and so ϕ should be similar to $\sin(2\theta)$.

To deduce what form ϕ should take we need to find the pressure p . To apply the boundary condition we first need to derive an expression for the pressure p . The stokes equation is:

$$\mathbf{0} = -\nabla p + \eta \nabla^2 \mathbf{u} \quad (2.14)$$

or in components:

$$0 = -\frac{\partial p}{\partial r} + \eta \left(\nabla^2 u - \frac{u}{r^2} - \frac{2}{r^2} \frac{\partial v}{\partial \theta} \right) \quad (2.15)$$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \eta \left(\nabla^2 v - \frac{v}{r^2} + \frac{2}{r^2} \frac{\partial u}{\partial \theta} \right) \quad (2.16)$$

where

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \quad (2.17)$$

Integrating equation (2.16) gives:

$$f_i(r) = -p + \eta \int \left(\frac{\partial}{\partial r} \left(r \frac{\partial v}{\partial r} \right) + \frac{1}{r} \frac{\partial^2 v}{\partial \theta^2} - \frac{v}{r} + \frac{2}{r} \frac{\partial u}{\partial \theta} \right) d\theta \quad (2.18)$$

Writing this in terms of the stream function ϕ we get

$$f_i(r) = -p + \eta \int \left(-\frac{\partial}{\partial r} \left(r \frac{\partial^2 \phi}{\partial r^2} \right) - \frac{1}{r} \frac{\partial^3 \phi}{\partial \theta^2 \partial r} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{2}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} \right) d\theta \quad (2.19)$$

Factoring out the $\frac{\partial}{\partial \theta}$ gives

$$f_i(r) = -p + \eta \int \left(-\frac{\partial}{\partial r} \left(r \frac{\partial^2 \phi}{\partial r^2} \right) + \frac{1}{r} \frac{\partial \phi}{\partial r} \right) d\theta + \eta \int \frac{\partial}{\partial \theta} \left(-\frac{1}{r} \frac{\partial^2 \phi}{\partial \theta \partial r} + \frac{2}{r^2} \frac{\partial \phi}{\partial \theta} \right) d\theta \quad (2.20)$$

$$f_i(r) = -p + \eta \int \left(-\frac{\partial}{\partial r} \left(r \frac{\partial^2 \phi}{\partial r^2} \right) + \frac{1}{r} \frac{\partial \phi}{\partial r} \right) d\theta + \eta \frac{\partial}{\partial \theta} \left(-\frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{2\phi}{r^2} \right) \quad (2.21)$$

The pressure must be 2π periodic so this suggests ϕ should be of the form $\phi_i = R_i(r) \sin(2\theta)$. To satisfy the Young Laplace equation we hope that the constant of integration satisfies this. Integrating (2.15) gives

$$g_i(\theta) = -p + \eta \int \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} - \frac{u}{r^2} - \frac{2}{r^2} \frac{\partial v}{\partial \theta} \right) dr \quad (2.22)$$

Writing this in terms of the stream function ϕ we get

$$g_i(\theta) = -p + \eta \int \left(\frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\partial^2 \phi}{\partial r \partial \theta} - \frac{1}{r} \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^3} \frac{\partial^3 \phi}{\partial \theta^3} - \frac{1}{3} \frac{\partial \phi}{\partial \theta} + \frac{2}{r^2} \frac{\partial^2 \phi}{\partial \theta \partial r} \right) dr \quad (2.23)$$

The biharmonic equation is

$$\nabla^4 \phi = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right)^2 \phi = 0 \quad (2.24)$$

Let

$$\mathcal{L} = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) \quad (2.25)$$

From prior observations we deduced that $\phi = R_i(r) \sin(2\theta)$

$$\mathcal{L}^2 [R_i(r) \sin(2\theta)] = 0 \quad (2.26)$$

So

$$\mathcal{L}^2 [R_i(r) \sin(2\theta)] = \mathcal{L} \left(\left(R'' + \frac{R'}{r} - \frac{4R}{r^2} \right) \sin(2\theta) \right) \quad (2.27)$$

To take the $\sin(2\theta)$ term out of the operator we evaluate the last term of $\mathcal{L}$ to get

$$\tilde{\mathcal{L}} \left(R'' + \frac{R'}{r} - \frac{4R}{r^2} \right) \sin(2\theta) = 0 \quad (2.28)$$

where

$$\tilde{\mathcal{L}} = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} - \frac{4}{r^2} \right) \quad (2.29)$$

Writing $R_i(r) = r^m$ gives

$$(m(m-1) + m - 4) \tilde{\mathcal{L}} (r^{m-2}) = 0 \quad (2.30)$$

$$(m+2)(m-2)(m(m-4)) = 0 \quad (2.31)$$

Hence $m = 2, -2, 4, 0$. So the general form of the solution is

$$R_i(r) = c + cr^2 + cr^{-2} + cr^4 \quad (2.32)$$

Applying far field conditions reduces this to

$$R_2(r) = c_3 + c_4 r^{-2} \quad (2.33)$$

For inside the drop we apply finite velocity conditions

$$R_1(r) = c_1 r^2 + c_2 r^4 \quad (2.34)$$

Hence we have:

$$\phi_1 = (c_1 r^2 + c_2 r^4) \sin(2\theta) \quad (2.35)$$

$$\phi_2 = (c_3 + c_4 r^{-2}) \sin(2\theta) \quad (2.36)$$

Now we have forms of ϕ_i we return to the momentum equation (2.21) and evaluate it with ϕ .

$$f_i(r) = -p - \frac{\eta_i}{2} \cos(2\theta) (-r R_i''' - R_i'' + 5R_i' r^{-1} - 8R_i r^{-2}) \quad (2.37)$$

Subbing in the values of ϕ_i gives

$$f_1(r) = -p + 12\eta_1 c_2 r^2 \cos(2\theta) \quad (2.38)$$

$$f_2(r) = -p + 4\eta_2 c_3 r^{-2} \cos(2\theta) \quad (2.39)$$

Now looking at (2.23) we get

$$g_i(\theta) = -p + \eta \cos(2\theta) \int (2R''r^{-1} + 2R'r^{-2} - 8Rr^{-3}) dr \quad (2.40)$$

$$g_1(\theta) = -p + 12\eta_1 c_2 r^2 \cos(2\theta) \quad (2.41)$$

$$g_2(\theta) = -p + 4\eta_2 c_3 r^{-2} \cos(2\theta) \quad (2.42)$$

So $f_1(r) = g_1(\theta)$ thus $f_1 = g_1 = A$ constant say. And $f_2(r) = g_2(\theta)$ so $f_2 = g_2 = B$ constant say. Imposing an ambient pressure p_0 say at $|r| \rightarrow \infty$ gives $B = -p_0$ Hence we have

$$p_1 = -A + 12\eta_1 c_2 r^2 \cos(2\theta) \quad (2.43)$$

$$p_2 = p_0 + 4\eta_2 c_3 r^{-2} \cos(2\theta) \quad (2.44)$$

In order to satisfy the magnetic Young-Laplace equation on the boundary we require $-p_0 - A - \frac{\mu_2 H_\infty B}{a^2} = \gamma \kappa = \frac{\gamma}{a}$, And so the pressures are

$$p_1 = p_0 + \frac{\gamma}{a} + \frac{\mu_2 H_\infty^2 (\mu_2 - \mu_1)}{\mu_2 + \mu_1} + 12c_2 \eta_1 r^2 \cos(2\theta) \quad (2.45)$$

$$p_2 = p_0 + 4c_3 \eta_2 r^{-2} \cos(2\theta) \quad (2.46)$$

We can now evaluate (2.11) as follows:

$$\begin{aligned} \frac{\gamma}{a} + \frac{\mu_2 H_\infty^2 (\mu_2 - \mu_1)}{\mu_2 + \mu_1} + (6\eta_1 c_2 r^2 + 2\eta_1 c_1 + 2\eta_2 c_4 r^{-4} - 6\eta_2 c_2 r^{-2}) \cos(2\theta) = \\ \frac{\gamma}{a} + \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)}{\mu_2 + \mu_1} \left(\frac{\mu_2 - \mu_1}{\mu_2 + \mu_1} \cos(2\theta) + 1 \right) \end{aligned} \quad (2.47)$$

Hence we get

$$6\eta_1 c_2 a^2 + 2\eta_1 c_1 + 2\eta_2 c_4 a^{-4} - 6\eta_2 c_2 a^{-2} = \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \quad (2.48)$$

Condition (2.9) becomes

$$\eta_2 (3c_4 a^{-4} + c_3 a^{-2}) - \eta_1 (3c_2 a^2 + c_1) = 0 \quad (2.49)$$

Continuity of fluid velocity (2.10) becomes

$$c_3 + c_4 a^{-2} - c_1 a^2 - c_2 a^4 = 0 \quad (2.50)$$

and

$$c_4 a^{-4} + c_1 + 2c_2 a^2 = 0 \quad (2.51)$$

Solving them we get

$$c_1 = -\frac{3(\eta_1 - 3\eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2}, \quad c_2 = \frac{(\eta_1 - 5\eta_2)}{16a^2 \eta_2 (\eta_1 - 2\eta_2)} \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \quad (2.52)$$

$$c_3 = -\frac{3a^2(\eta_1 - \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2}, \quad c_4 = \frac{a^4(\eta_1 + \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \quad (2.53)$$

And so the solution is

$$\phi_1 = \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(-\frac{3(\eta_1 - 3\eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r^2 + \frac{(\eta_1 - 5\eta_2)}{16a^2\eta_2(\eta_1 - 2\eta_2)} r^4 \right) \sin(2\theta) \quad (2.54)$$

and

$$\phi_2 = \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(-\frac{3a^2(\eta_1 - \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} + \frac{a^4(\eta_1 + \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r^{-2} \right) \sin(2\theta) \quad (2.55)$$

Recovering the velocities we get

$$u_1 = \frac{2H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(-\frac{3(\eta_1 - 3\eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r + \frac{(\eta_1 - 5\eta_2)}{16a^2\eta_2(\eta_1 - 2\eta_2)} r^3 \right) \cos(2\theta) \quad (2.56)$$

$$u_2 = \frac{2H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(-\frac{3a^2(\eta_1 - \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r^{-1} + \frac{a^4(\eta_1 + \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r^{-3} \right) \cos(2\theta) \quad (2.57)$$

$$v_1 = \frac{-H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(-\frac{6(\eta_1 - 3\eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r + \frac{4(\eta_1 - 5\eta_2)}{16a^2\eta_2(\eta_1 - 2\eta_2)} r^3 \right) \sin(2\theta) \quad (2.58)$$

$$v_2 = \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{(\mu_2 + \mu_1)^2} \left(\frac{2a^4(\eta_1 + \eta_2)}{16\eta_2(\eta_1 - 2\eta_2)} r^{-3} \right) \sin(2\theta) \quad (2.59)$$

or

$$u_1 = \frac{H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \left(-3(\eta_1 - 3\eta_2)r + (\eta_1 - 5\eta_2) \frac{r^3}{a^2} \right) \cos(2\theta) \quad (2.60)$$

$$u_2 = \frac{a^2 H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \left(-3(\eta_1 - \eta_2)r^{-1} + a^2(\eta_1 + \eta_2)r^{-3} \right) \cos(2\theta) \quad (2.61)$$

$$v_1 = \frac{-H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \left(-3(\eta_1 - 3\eta_2)r + 2(\eta_1 - 5\eta_2) \frac{r^3}{a^2} \right) \sin(2\theta) \quad (2.62)$$

$$v_2 = \frac{a^4 H_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \left((\eta_1 + \eta_2)r^{-3} \right) \sin(2\theta) \quad (2.63)$$

We can deduce what will happen by looking at where the boundary wants to move to. By setting $r = a$ we get

$$u_1|_{r=a} = \frac{-aH_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2} \cos(2\theta) \quad (2.64)$$

$$u_2|_{r=a} = \frac{-aH_\infty^2 \mu_2 (\mu_2 - \mu_1)^2}{8(\mu_2 + \mu_1)^2 \eta_2} \cos(2\theta) \quad (2.65)$$

$$v_1|_{r=a} = \frac{aH_\infty^2 \mu_2 (\mu_2 - \mu_1)^2 (\eta_1 + \eta_2)}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \sin(2\theta) \quad (2.66)$$

$$v_2|_{r=a} = \frac{aH_\infty^2 \mu_2 (\mu_2 - \mu_1)^2 (\eta_1 + \eta_2)}{8(\mu_2 + \mu_1)^2 \eta_2 (\eta_1 - 2\eta_2)} \sin(2\theta) \quad (2.67)$$

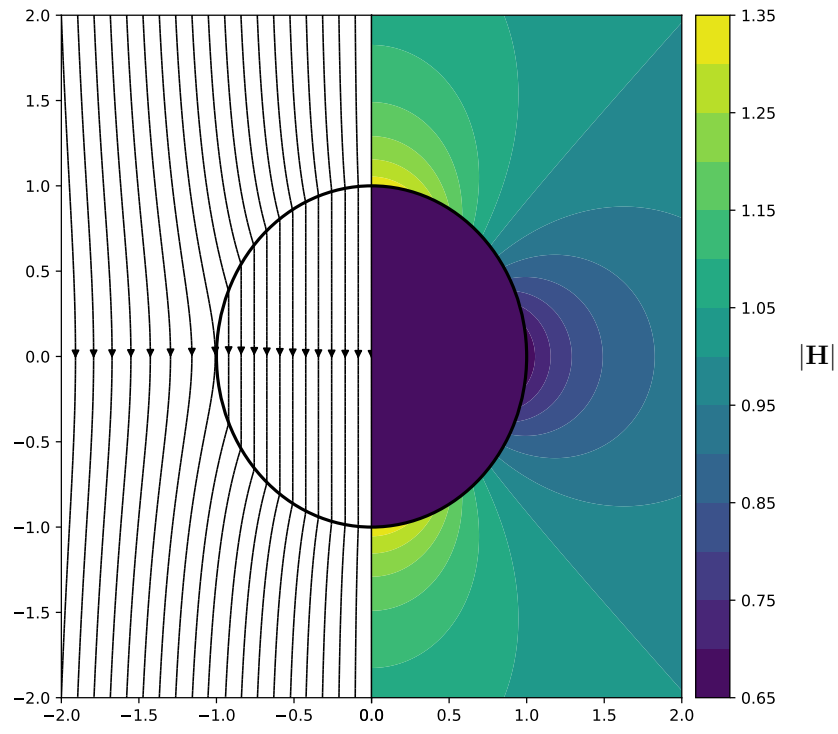


Figure 1: Streamlines of magnetic field (left), magnitude of magnetic field (right)

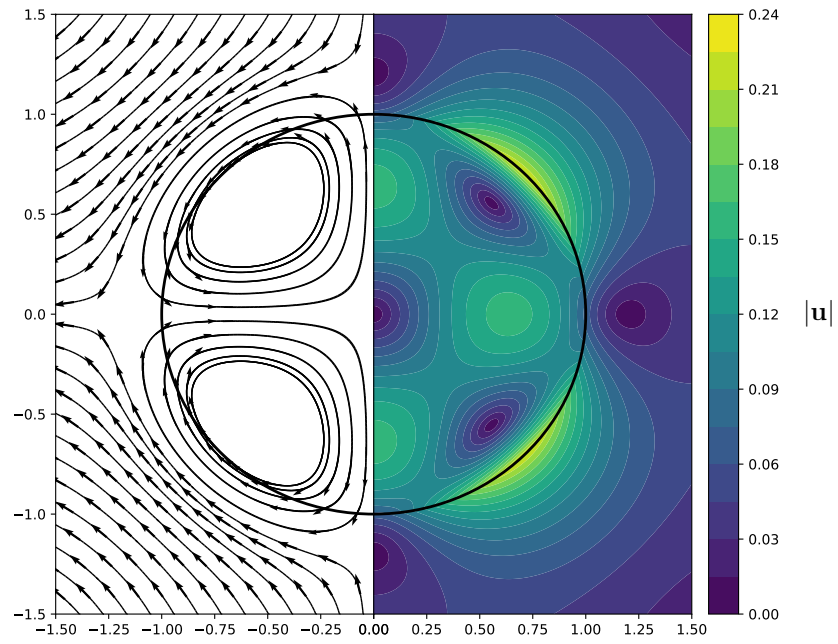


Figure 2: Streamlines of instantaneous fluid motion (left), magnitude of fluid motion (right)

3 Perturbed boundary: WRONG APPROACH

Let the boundary be defined by $r = b(1 + \epsilon \cos(n\theta)) = f(\theta)$. The normal is defined as $\mathbf{n} = \frac{\nabla(r-f)}{|\nabla(r-f)|}$. Hence the normal and tangential directions are

$$\mathbf{n} = \frac{f\mathbf{e}_r - f'\mathbf{e}_\theta}{M}, \quad \mathbf{t} = \frac{f'\mathbf{e}_r + f\mathbf{e}_\theta}{M}$$

respectively. The jump boundary conditions are $[\mathbf{B} \cdot \mathbf{n}] = 0$ and $[\mathbf{H} \cdot \mathbf{t}] = 0$. On the outer boundary we write $[\mathbf{B} \cdot \mathbf{n}] = 0$ as $\mathbf{B}_2 \cdot \mathbf{n} = \mathbf{B}_3 \cdot \mathbf{n}$ which is equivalent to

$$f^2 \left(\mu_0 \frac{\partial \psi_2}{\partial r} - \mu_b \frac{\partial \psi_3}{\partial r} \right) = f' \left(\mu_0 \frac{\partial \psi_2}{\partial \theta} - \mu_b \frac{\partial \psi_3}{\partial \theta} \right)$$

Taylor expanding about $r = b(1 + \epsilon \cos(n\theta))$ and writing $\psi = \bar{\psi} + \epsilon \psi^* + O(\epsilon^2)$ where $\bar{\psi}$ has been found to be

$$\bar{\psi} = \begin{cases} c_1 r \sin \theta & r < a \\ (c_2 + c_3/r) \sin \theta & a < r < b \\ (H_\infty r + c_5/r) \sin \theta & r < b \end{cases}$$

for constants c_i . We derive the first boundary condition to be

$$\frac{\partial}{\partial r}(\mu_0 \psi_2^* - \mu_b \psi_3^*) = \frac{n \sin(n\theta)}{b} \frac{\partial}{\partial \theta}(\mu_b \bar{\psi}_3 - \mu_0 \bar{\psi}_2) + b \cos(n\theta) \frac{\partial^2}{\partial r^2}(\mu_b \bar{\psi}_3 - \mu_0 \bar{\psi}_2)$$

The jump condition $\mathbf{H}_2 \cdot \mathbf{t} = \mathbf{H}_3 \cdot \mathbf{t}$ equates to

$$f' \left(\frac{\psi_3}{\partial r} - \frac{\psi_2}{\partial r} \right) = \frac{\partial \psi_3}{\partial \theta} - \frac{\partial \psi_2}{\partial \theta}$$

Taylor expanding about $r = b(1 + \epsilon \cos(n\theta))$ and writing $\psi = \bar{\psi} + \epsilon \psi^* + O(\epsilon^2)$ we derive the second boundary condition

$$\frac{\partial}{\partial \theta}(\psi_2^* - \psi_3^*) = n \sin(n\theta) \frac{\partial}{\partial r}(\bar{\psi}_2 - \bar{\psi}_3)$$

A similar thing can be done for the boundary $r = a(1 + \epsilon g(\theta))$. For $[\mathbf{B} \cdot \mathbf{n}] = 0$ we get

$$\frac{\partial}{\partial r}(\mu_a \psi_1^* - \mu_0 \psi_2^*) = \frac{g'}{a} \frac{\partial}{\partial \theta}(\mu_a \bar{\psi}_1 - \mu_0 \bar{\psi}_2) + a g \frac{\partial^2}{\partial r^2}(\mu_0 \bar{\psi}_2 - \mu_a \bar{\psi}_1)$$

For $[\mathbf{H} \cdot \mathbf{t}] = 0$ we get

$$\frac{\partial}{\partial \theta}(\psi_1^* - \psi_2^*) = a g' \frac{\partial}{\partial r}(\bar{\psi}_1 - \bar{\psi}_2)$$

Hence the boundary conditions are as follows:

$$\mathbf{B}_2 \cdot \mathbf{n} = \mathbf{B}_3 \cdot \mathbf{n}:$$

$$\frac{\partial}{\partial r}(\mu_0 \psi_2^* - \mu_b \psi_3^*) = \frac{n \sin(n\theta)}{b} \frac{\partial}{\partial \theta}(\mu_b \bar{\psi}_3 - \mu_0 \bar{\psi}_2) + b \cos(n\theta) \frac{\partial^2}{\partial r^2}(\mu_b \bar{\psi}_3 - \mu_0 \bar{\psi}_2)$$

$$\mathbf{H}_2 \cdot \mathbf{t} = \mathbf{H}_3 \cdot \mathbf{t}:$$

$$\frac{\partial}{\partial \theta}(\psi_2^* - \psi_3^*) = n \sin(n\theta) \frac{\partial}{\partial r}(\bar{\psi}_2 - \bar{\psi}_3)$$

$$\mathbf{B}_1 \cdot \mathbf{n} = \mathbf{B}_2 \cdot \mathbf{n}:$$

$$\frac{\partial}{\partial r}(\mu_d \psi_1^* - \mu_0 \psi_2^*) = \frac{g'}{a} \frac{\partial}{\partial \theta}(\mu_d \bar{\psi}_1 - \mu_0 \bar{\psi}_2) + ag \frac{\partial^2}{\partial r^2}(\mu_0 \bar{\psi}_2 - \mu_d \bar{\psi}_1)$$

$$\mathbf{H}_1 \cdot \mathbf{t} = \mathbf{H}_2 \cdot \mathbf{t}:$$

$$\frac{\partial}{\partial \theta}(\psi_1^* - \psi_2^*) = ag' \frac{\partial}{\partial r}(\bar{\psi}_1 - \bar{\psi}_2)$$

$\bar{\psi}$ is known and can be written in each region as $\bar{\psi}_i = \bar{R}_i \sin \theta$ for each region $i = 1, 2, 3$. We also write the perturbation quantity ψ^* as $\psi^* = R(r)T(\theta)$ as ψ^* satisfies $\nabla^2 \psi^* = 0$.

Hence we can write

$$\mathbf{B}_2 \cdot \mathbf{n} = \mathbf{B}_3 \cdot \mathbf{n}:$$

$$(\mu_0 R'_2 - \mu_b R'_3)T = \frac{n \sin(n\theta) \cos \theta}{b}(\mu_b \bar{R}_3 - \mu_0 \bar{R}_2) + b \cos(n\theta) \sin \theta(\mu_b \bar{R}_3'' - \mu_0 \bar{R}_2'')$$

$$\mathbf{H}_2 \cdot \mathbf{t} = \mathbf{H}_3 \cdot \mathbf{t}:$$

$$(R_2 - R_3)T' = n \sin(n\theta) \sin \theta(\bar{R}_2 - \bar{R}_3)$$

$$\mathbf{B}_1 \cdot \mathbf{n} = \mathbf{B}_2 \cdot \mathbf{n}:$$

$$(\mu_d R'_1 - \mu_0 R'_2)T = \frac{g'}{a} \cos \theta(\mu_d \bar{R}_1 - \mu_0 \bar{R}_2) + ag \sin \theta(\mu_0 \bar{R}_2'' - \mu_d \bar{R}_1'')$$

$$\mathbf{H}_1 \cdot \mathbf{t} = \mathbf{H}_2 \cdot \mathbf{t}:$$

$$(R_1 - R_2)T' = ag' \sin \theta(\bar{R}_1 - \bar{R}_2)$$

We require $g(\theta) = X_n \cos \theta$ for constants X_n .

Using the trig identities:

$$A \sin(n\theta) \cos \theta + B \cos(n\theta) \sin \theta = \frac{1}{2}(A + B) \sin((n+1)\theta) + \frac{1}{2}(A - B) \sin((n-1)\theta)$$

and

$$A \sin(n\theta) \sin \theta = \frac{A}{2}(\cos((n-1)\theta) - \cos((n+1)\theta))$$

Hence:

$$\mathbf{B}_2 \cdot \mathbf{n} = \mathbf{B}_3 \cdot \mathbf{n}:$$

$$\begin{aligned} (\mu_0 R'_2 - \mu_b R'_3)T &= \frac{1}{2} \left[\left(\frac{n}{b}(\mu_b \bar{R}_3 - \mu_0 \bar{R}_2) + b(\mu_b \bar{R}_3'' - \mu_0 \bar{R}_2'') \right) \sin((n+1)\theta) \right. \\ &\quad \left. + \frac{1}{2} \left[\frac{n}{b}(\mu_b \bar{R}_3 - \mu_0 \bar{R}_2) - b(\mu_b \bar{R}_3'' - \mu_0 \bar{R}_2'') \right] \sin((n-1)\theta) \right] \end{aligned}$$

$$\mathbf{H}_2 \cdot \mathbf{t} = \mathbf{H}_3 \cdot \mathbf{t}:$$

$$(R_2 - R_3)T' = \frac{n}{2}(\bar{R}_2 - \bar{R}_3)(\cos((n-1)\theta) - \cos((n+1)\theta))$$

$$\mathbf{B}_1 \cdot \mathbf{n} = \mathbf{B}_2 \cdot \mathbf{n}:$$

$$\begin{aligned} (\mu_d R'_1 - \mu_0 R'_2)T &= \frac{1}{2} \left[\frac{-nX_n}{a} (\mu_d \bar{R}_1 - \mu_0 \bar{R}_2) + a(\mu_0 \bar{R}_2'' - \mu_d \bar{R}_1'') \right] \sin((n+1)\theta) + \\ &\quad \frac{1}{2} \left[\frac{-nX_n}{a} (\mu_d \bar{R}_1 - \mu_0 \bar{R}_2) - a(\mu_0 \bar{R}_2'' - \mu_d \bar{R}_1'') \right] \sin((n-1)\theta) \end{aligned}$$

$$\mathbf{H}_1 \cdot \mathbf{t} = \mathbf{H}_2 \cdot \mathbf{t}:$$

$$(R_1 - R_2)T' = \frac{-nX_n a}{2} (\bar{R}_1 - \bar{R}_2) (\cos((n-1)\theta) - \cos((n+1)\theta))$$

We infer that the separation constant $\lambda = n+1, n-1$. Hence $T(\theta) = \alpha_1 \sin((n+1)\theta) + \alpha_2 \sin((n-1)\theta)$.

Equating coefficients we get

$$\begin{aligned} (\mu_0 R'_2 - \mu_b R'_3)\alpha_1 &= \frac{1}{2} \left[\left(\frac{n}{b}(\mu_b \bar{R}_3 - \mu_0 \bar{R}_2) + b(\mu_b \bar{R}_3'' - \mu_0 \bar{R}_2'')\right) \right] \\ (\mu_0 R'_2 - \mu_b R'_3)\alpha_2 &= \frac{1}{2} \left[\left(\frac{n}{b}(\mu_b \bar{R}_3 - \mu_0 \bar{R}_2) - b(\mu_b \bar{R}_3'' - \mu_0 \bar{R}_2'')\right) \right] \\ (R_2 - R_3)(n+1)\alpha_1 &= \frac{-n}{2} (\bar{R}_2 - \bar{R}_3) \\ (R_2 - R_3)(n-1)\alpha_2 &= \frac{n}{2} (\bar{R}_2 - \bar{R}_3) \\ (\mu_d R'_1 - \mu_0 R'_2)\alpha_1 &= \frac{X_n}{2} \left[\frac{-n}{a} (\mu_d \bar{R}_1 - \mu_0 \bar{R}_2) + a(\mu_0 \bar{R}_2'' - \mu_d \bar{R}_1'') \right] \\ (\mu_d R'_1 - \mu_0 R'_2)\alpha_2 &= \frac{X_n}{2} \left[\frac{-n}{a} (\mu_d \bar{R}_1 - \mu_0 \bar{R}_2) - a(\mu_0 \bar{R}_2'' - \mu_d \bar{R}_1'') \right] \\ (R_1 - R_2)(n+1)\alpha_1 &= \frac{nX_n a}{2} (\bar{R}_1 - \bar{R}_2) \\ (R_1 - R_2)(n-1)\alpha_2 &= \frac{-nX_n a}{2} (\bar{R}_1 - \bar{R}_2) \end{aligned}$$